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30. Do you think that heavy water can be used for drinking purposes ?

31. What is water-gas shift reaction ?

32. Justify the position of hydrogen in the periodic table ?

33. What are isotopes? Write the names of isotopes of hydrogen.

34. Give the uses of heavy water.

35. Explain the exchange reactions of deuterium.

36. How do you convert parahydrogen into ortho hydrogen ?

37. Mention the uses of deuterium.

38. Explain preparation of hydrogen using electrolysis.

39. A groups metal (A) which is present in common salt reacts with (B) to give compound (C) in which hydrogen is present in –1 oxidation state. (B) on reaction with a gas (C) to give universal solvent (D). The compound (D) on reacts with (A) to give (B), a strong base. Identify A, B, C, D and E. Explain the reactions.

40. An isotope of hydrogen (A) reacts with diatomic molecule of element which occupies group number 16 and period number 2 to give compound (B) is used as a modulator in nuclear reaction. (A) adds on to a compound (C), which has the molecular formula C_3H_6 to give (D). Identify A, B, C and D.

41. NH_3 has exceptionally high melting point and boiling point as compared to those of the hydrides of the remaining element of group 15 - Explain.

42. Why interstitial hydrides have a lower density than the parent metal.

43. How do you expect the metallic hydrides to be useful for hydrogen storage ?

44. Arrange NH_3, H_2O and HF in the order of increasing magnitude of hydrogen banding and explain the basis for your arrangement.

45. Compare the structures of H_2O and H_2O_2.